quency of 353 cm^{-1} and estimated the heat of formation as 212.3 kcal/mol. His calculation was based on the standard rigid-rotor harmonic-oscillator model. In 1959 Pitzer and Clementi (ref. 20) corrected the multiplicity of the ground state to 1 and recomputed the free-energy function and the enthalpy using a bending frequency of 550 cm^{-1}. However, in 1965 Gausset, Herzberg, Lagerqvist, and Rosen (ref. 3) published a detailed spectroscopic analysis of C_3, concluding that the bending frequency of the ground electronic state was of the order of 65 cm^{-1} for the first few levels. This low bending frequency created a difficulty in calculating the thermal functions, for as shown by Robiette and Strauss (ref. 28), the assumption of a harmonic oscillator with the lower value of bending frequency causes the calculated entropy to be too high compared with that obtained from the temperature measurements. For example, Thorn and Winslow (ref. 29) calculated a value of entropy S_{2400} of 77.41 eu from their vapor-pressure measurement, and Robiette and Strauss (ref. 28) interpreted the pressure measurements of Drowart et al. (ref. 13) to yield S_{2400} = 76.1 eu. Robiette and Strauss showed, however, that at 2400 K a harmonic-oscillator calculation with a bending frequency of 63 cm^{-1} yields an entropy of 85.56 eu. They showed that the entropy can be reduced by about 3 eu by taking the quartic anharmonicity into account.

Strauss and Thiele (ref. 30) calculated the thermal properties of C_3 using the spectroscopic data of Gausset et al. (ref. 3) and accounting for rotation-vibration interaction. However, as they pointed out, only the first few levels of the bending vibration have been measured, so that the potential function must be based on a very limited set of energy levels. They show that inclusion of the quartic anharmonicity lowers the entropy at 2400 K by about 2.8 eu, and inclusion of the vibration-rotation coupling lowers it by another 1.1 eu, to a value of 81.4. Further, they demonstrated that even if the potential increased drastically above a bending angle of 0.4π, the calculated entropy cannot be lowered below S_{2400} = 79.8 eu. It appears therefore, that the best calculation method cannot compute entropy values as low as those inferred from the experiments of references 13 and 29.

The JANAF tables (ref. 26) adopted a potential function which makes use of the vibrational frequencies measured by Gausset et al. (ref. 3) for the low levels and then steepens to fit a vibrational frequency interval of 650 cm^{-1} for higher levels. The value of 650 cm^{-1} is admitted to be arbitrary and was selected to yield functions which approximate experimental data. Their value of entropy at 2400 K is S_{2400} = 78.4 eu.

Hansen and Pearson (ref. 31) used a quadratically perturbed square well potential in a quantum calculation to approximate the spectroscopic data of C_3. The maximum molecular bending at the potential wall was 64° for this model, which is in close agreement with Strauss and Thiele's approximate model (ref. 30), which steepened the potential at a bending angle of 0.4π rad, or 72°. From their analysis, Hansen and Pearson calculated an entropy at 2400 K of S_{2400} = 79.68 eu, and concluded that their limiting calculation thereby supports the calculation of Strauss and Thiele.

Lee and Sanborn (ref. 15) also calculated the thermal properties of C_3 by developing a special potential model. Their anharmonic model for the lower

vibrational levels fits the data of Gausset et al. (ref. 3), and for higher vibrational levels the term difference approaches 308 cm^{-1}, the bending frequency of the first excited state. They also included the electronic contribution to the partition function, which had been neglected in the other treatments. However, they used a rigid-rotor assumption and thus neglected vibration-rotation interaction. Lee and Sanborn showed that their results are in close agreement with those of Strauss and Thiele (ref. 30) when they omitted the electronic contribution. They used the electronic energy levels given by Pitzer and Clementi (ref. 20) in computing the electronic contribution to the partition function.

Figure 17 compares the specific heat at constant pressure, the enthalpy, the entropy, and the Gibbs energy function as calculated by references 15, 26, 30, and 31. The differences between the calculations are most accentuated in c_p, but are appreciable in the other functions as well. The Gibbs functions as computed by Lee and Sanborn are seen to be quite close to those by Strauss and Thiele throughout the temperature range. About half the difference between these curves may be attributed to the electronic contribution which Lee and Sanborn included.

The calculations of Strauss and Thiele (ref. 30) would appear to be the most rigorous treatment of the C_3 molecule with the spectroscopic data available, except for omission of the electronic contribution to the partition function. Since the results of Lee and Sanborn are in acceptable agreement with the results of Strauss and Thiele when the electronic contribution is omitted, their thermal functions (ref. 15) seem to be the best choice. Lee and Sanborn also presented thermal functions for C_4 through C_7. In addition, the properties for C_1 and C_2 were extended to 10 000 K. Thus a consistent set of properties for the carbon species up to C_7 are available in reference 15 for carbon vapor calculations.

For the reasons stated, the equilibrium calculations of this study used curve fits to the calculated data of Lee and Sanborn for the thermodynamic functions of C_3.

One further point of interest may be noted regarding the evaluation of the various calculations of thermal properties. The most nagging problem has been the fact that each of the calculations yielded higher entropy values than those measured by Drowart et al. (ref. 13) and Thorn and Winslow (ref. 29), and resolving the low bending frequency with the experimental entropy values seemed impossible. In 1972, however, Wachi and Gilmartin (ref. 32) made new vapor-pressure measurements with a calibrated mass spectrometer and a Knudsen effusion cell. They quoted experimental entropies at 2400 and 2700 K as $S_{2400} = 81.47$ eu and $S_{2700} = 82.91$ eu. These values are shown in figure 17(c) and are in excellent agreement with the calculation of Strauss and Thiele (ref. 30) and in acceptable agreement with the calculation of Lee and Sanborn (ref. 15). Thus, if one assumes that the measurements of Wachi and Gilmartin are superior to the earlier ones, the discrepancy between theory and experiment is resolved on this point.